

Computational Thinking

Discrete Mathematics

Number Theory

Topic 00 : Module Introduction

Logic

Lecture 01 : Module Overview

Dr Kieran Murphy 

Computing and Mathematics, SETU (Waterford).
(kieran.murphy@setu.ie)

Graphs and
Networks

Autumn Semester, 2026/27

Collections

Outline

- Motivation and aim of this module.
- Administration trivia — Contact hours, Assessment structure, ...
- Resources

Enumeration

Relations & Functions

Outline

1. Module Introduction	2
1.1. What is the Aim of this Module?	3
1.2. Why Study Discrete Mathematics?	12
1.3. How will the Module be Delivered? Assessed?	14
1.4. Who is delivering this module?	17
1.5. When will the module be delivered?	18
2. Resources	18
2.1. Moodle	19
2.2. Github	20
2.3. Slack	21
2.4. Colab	22
2.5. PyTutor	23
3. Final Comments	25

What?

(Aim)

Aim, as per the Module Descriptor ...

This module provides a solid foundation of selected topics in discrete mathematics related to computing and information sciences. The topics are covered in an elementary manner in order to reinforce understanding of concepts and improving algebraic problem-solving skills so that the student can effectively proceed with their study of a degree programme in computing.

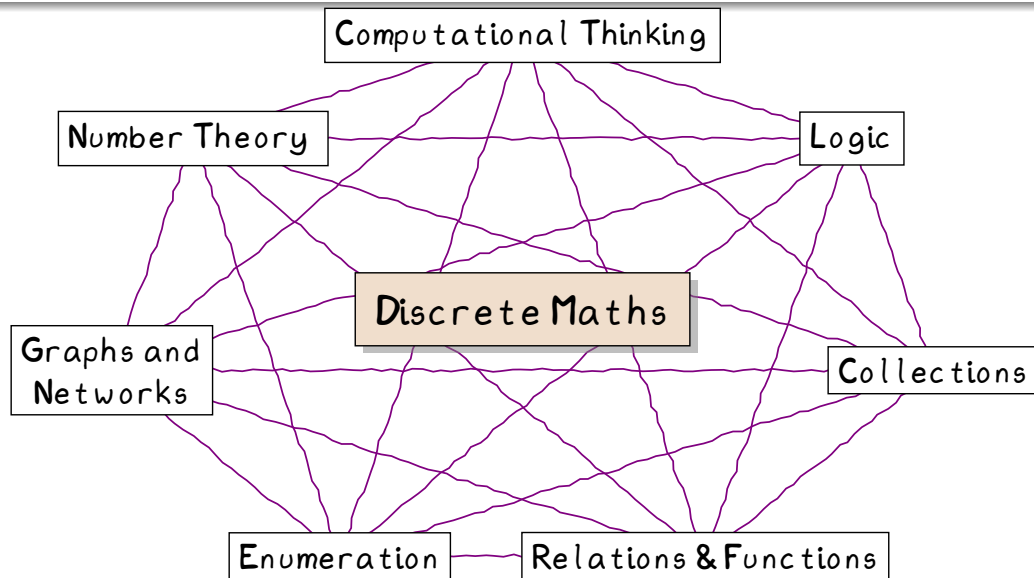
Translation (Informal Aims)

- 1 Reason logically — aim for precision and correctness over speed.
- 2 Develop and manipulate theoretical models
— a **set** is a collection of things, a **relation** is a collection of pairs of things, a **graph** is a collection of things with pairwise connections, etc..
- 3 Translate

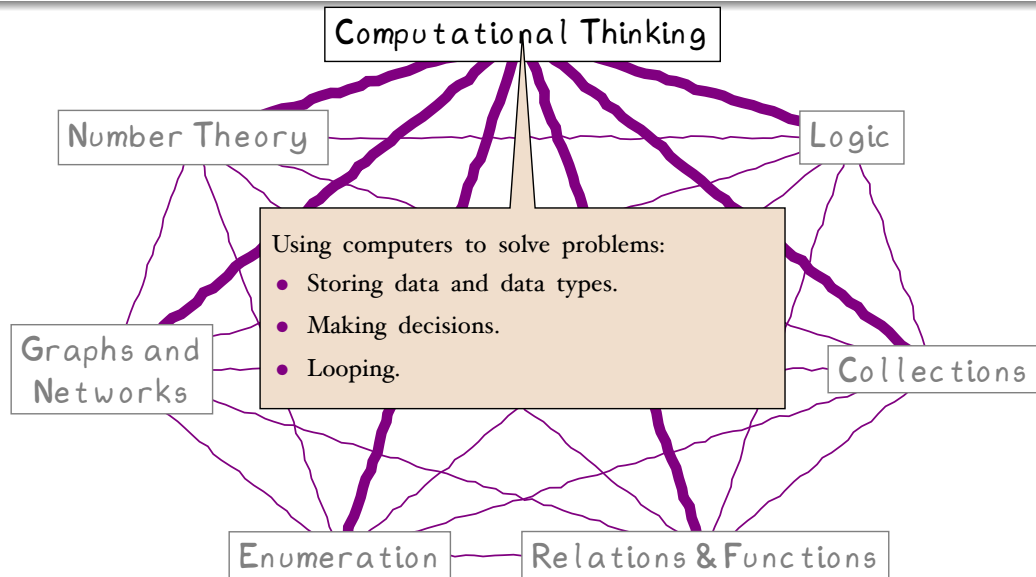
computing concepts/implementation (Python) $\leftrightarrow$ theoretical models (mathematics).

What?

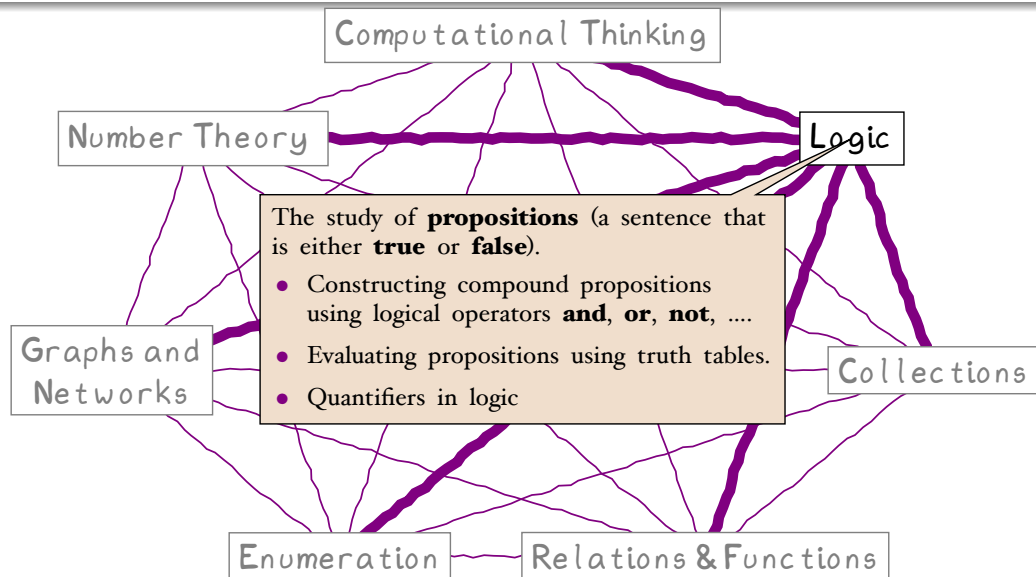
(Discrete Mathematics)



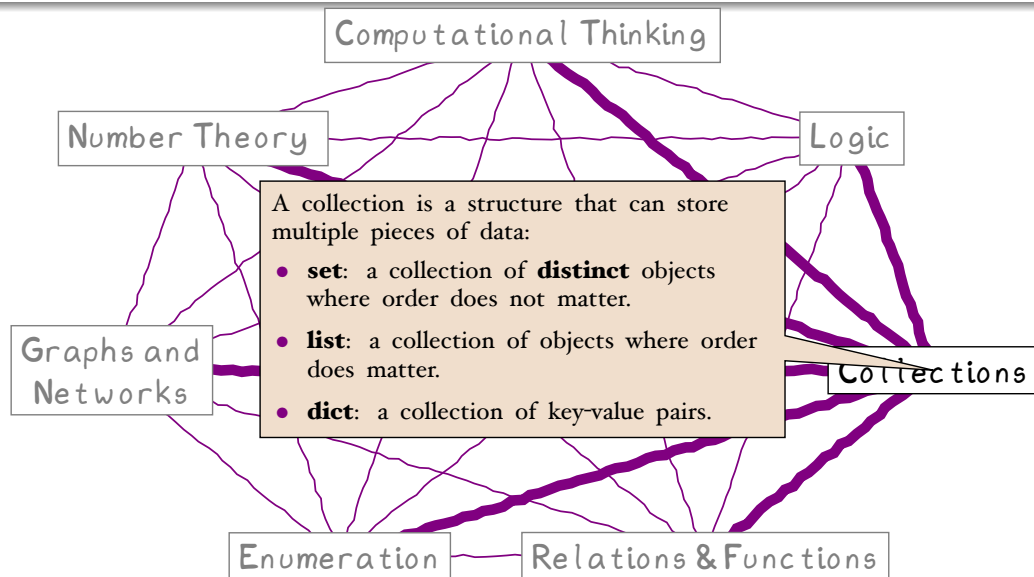
What?



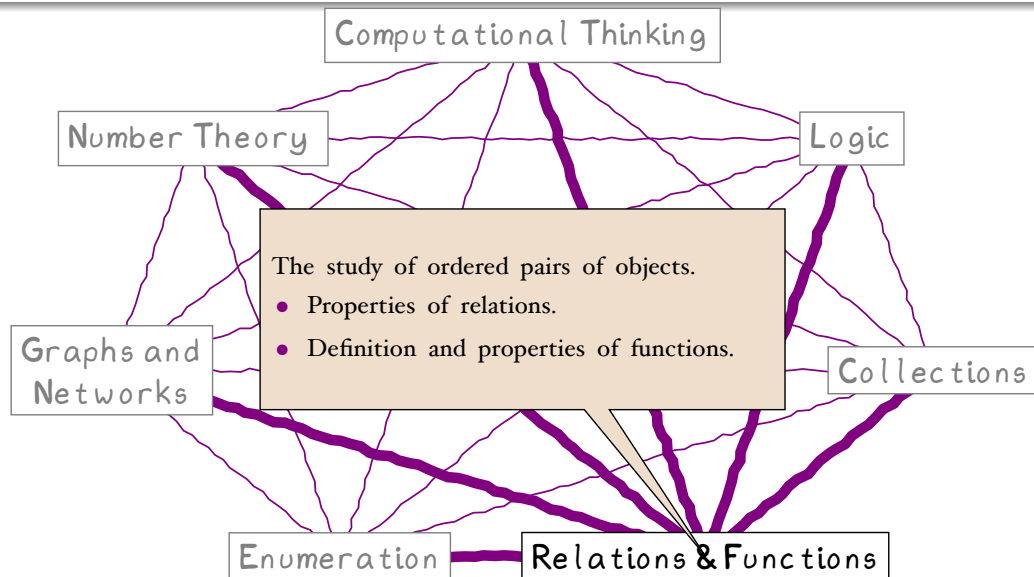
What?



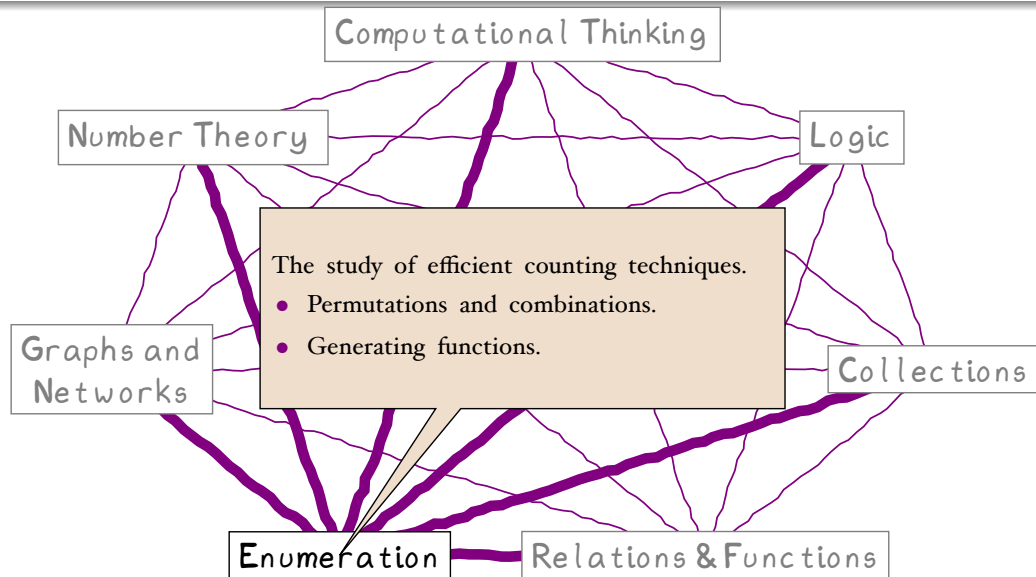
What?



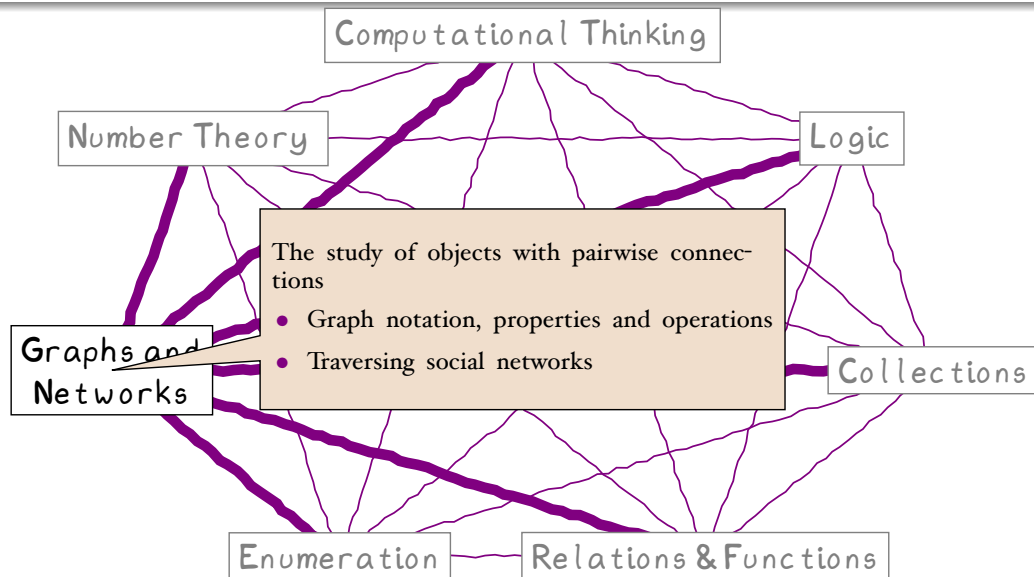
What?



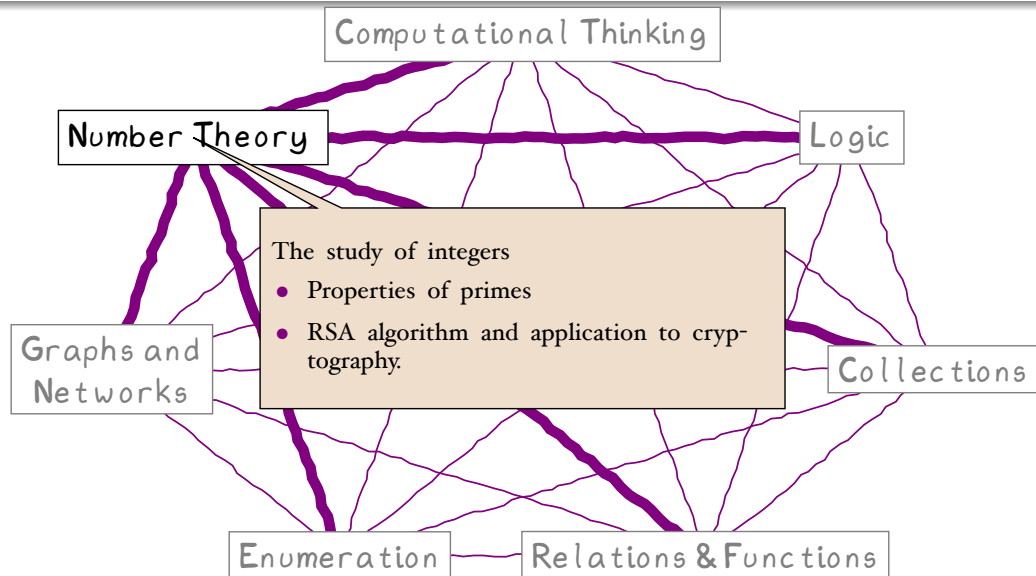
What?



What?



What?



Why?

Many, many reasons ... picking one* ...

AI / Machine learning is the future of computing

Discrete mathematics is the core of machine learning.

Why?

Many, many reasons ... picking one* ...

AI / Machine learning is the future of computing

Discrete mathematics is the core of machine learning.

Why?

Many, many reasons ... picking one* ...

AI / Machine learning is the future of computing

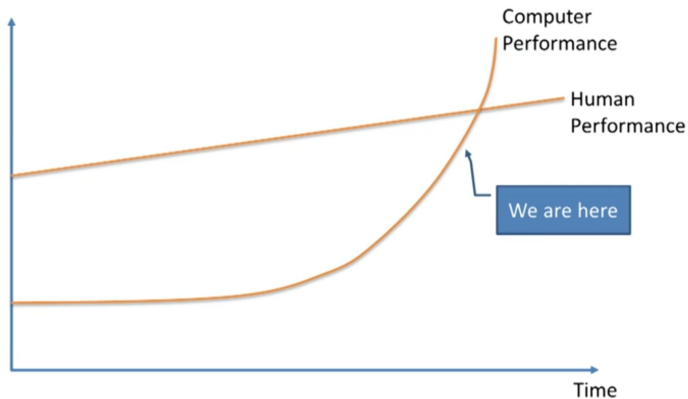
Discrete mathematics is the core of machine learning.

Why?

Many, many reasons ... picking one* ...

AI / Machine learning is the future of computing

Discrete mathematics is the core of machine learning.



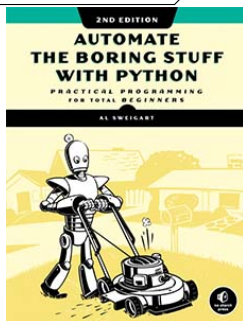
Why?

(Aim)

Computers are faster and more accurate* than us, they don't get board/distracted†

⇒ Get computers to do the hard (and/or boring) stuff

See Al Sweigart's books



Or our view on life



Why have a dog and bark?

Exception to this are the generative text models (chatGPT, Bard, ...) where accuracy is replaced by "most likely/probable".

†Well, if you don't count Teslas, in FSD mode, driving over pedestrians ...

Aside: Speaking of Elon ... AI and all that ...

List of predictions for autonomous Tesla vehicles by Elon Musk

From Wikipedia, the free encyclopedia

Date ↕	Prediction ↕	Quote	Met ↕
Sep 2013	2016	"We should be able to do 90 percent of miles driven [autonomously] within three years."	✗ No
Jun 2014	2015	"I am confident that in less than a year you will be able to go from highway on ramp to highway exits without touching any controls."	✗ No
Oct 2014	2015	"A Tesla car next year will probably be 90-percent capable of autopilot. Like, so 90 percent of your miles can be on auto. For sure highway travel."	AP/HW1 released Oct 2015 for highways^[7]
Oct 2014	2023	"I think we'll be able to achieve true autonomous driving, where you could literally get in the car, go to sleep and wake up at your destination,"	✗ No
Oct 2015	2018	"From a technology standpoint, Tesla will have a car that can do full autonomy in about three years, maybe a bit sooner."	✗ No

Dec 2015	2018	"We're going to end up with complete autonomy, and I think we will have complete autonomy in approximately two years."	✗ No
Jan 2016	2018	"Ultimately you'll be able to summon your car anywhere ... your car can get to you. I think that within two years, you'll be able to summon your car from across the country. It will meet you wherever your phone is ... and it will just automatically charge itself along the entire journey."	✗ No
Jun 2016	2019	"I consider autonomous driving to be a basically solved problem. ... We're less than two years away from complete autonomy. Regulators however will take at least another year; they'll want to see billions of miles of	✗ No

How?

(Contact Hours)

⚙️ Three lectures per week

- ⚙️ Cover concepts, definitions, examples, etc.

- ⚙️ BUT feel free to stop me and ask questions at any point.

- ⚠️ **You should print out notes in advance of lecture, or have access to them during class.**

- ⚙️ Ideally you skimmed over the notes in advance of lecture.

- ⚠️ **Take notes during lectures.**

⚙️ One tutorial per week

- ⚙️ Review of exercises based on the material covered in the lectures.

- ⚠️ **You need to have printout of tutorial sheets in advance of lecture.**

- ⚙️ Ideally you have attempted/completed some/all questions in advance of tutorial and you are just attending the tutorials to show off.

- ⚙️ Online quiz for self review at end of each topic.

⚙️ One practical per week

- ⚙️ Using Python (via *colab notebooks*) to demonstrate implementation details of discrete mathematics concepts.

- ⚙️ Introduce programming in Python — never have too much programming.

- ⚠️ **You need to upload notebook by end of week (Saturday 11:00pm).**

How?

(Contact Hours)

⚙️ Three lectures per week

- ⚙️ Cover concepts, definitions, examples, etc.

- ⚙️ BUT feel free to stop me and ask questions at any point.

- ⚠️ **You should print out notes in advance of lecture, or have access to them during class.**

- ⚙️ Ideally you skimmed over the notes in advance of lecture.

- ⚠️ **Take notes during lectures.**

⚙️ One tutorial per week

- ⚙️ Review of exercises based on the material covered in the lectures.

- ⚠️ **You need to have printout of tutorial sheets in advance of lecture.**

- ⚙️ Ideally you have attempted/completed some/all questions in advance of tutorial and you are just attending the tutorials to show off.

- ⚙️ Online quiz for self review at end of each topic.

⚙️ One practical per week

- ⚙️ Using Python (via *colab notebooks*) to demonstrate implementation details of discrete mathematics concepts.

- ⚙️ Introduce programming in Python — never have too much programming.

- ⚠️ **You need to upload notebook by end of week (Saturday 11:00pm).**

How?

(Contact Hours)

⚙️ Three lectures per week

- ⚙️ Cover concepts, definitions, examples, etc.

- ⚙️ BUT feel free to stop me and ask questions at any point.

- ⚠️ **You should print out notes in advance of lecture, or have access to them during class.**

- ⚙️ Ideally you skimmed over the notes in advance of lecture.

- ⚠️ **Take notes during lectures.**

⚙️ One tutorial per week

- ⚙️ Review of exercises based on the material covered in the lectures.

- ⚠️ **You need to have printout of tutorial sheets in advance of lecture.**

- ⚙️ Ideally you have attempted/completed some/all questions in advance of tutorial and you are just attending the tutorials to show off.

- ⚙️ Online quiz for self review at end of each topic.

⚙️ One practical per week

- ⚙️ Using Python (via *colab notebooks*) to demonstrate implementation details of discrete mathematics concepts.

- ⚙️ Introduce programming in Python — never have too much programming.

- ⚠️ **You need to upload notebook by end of week (Saturday 11:00pm).**

How?

(Assessment Structure)

75% End of Semester Exam

Current plan (this is subject to change so ask about this in week 10!)

- 4 questions (typically 3–5 parts per question. Answer all questions (i.e. no choice).
- Tend not to have question per topic.
- 80% same material as last year — but there may still be some differences in format/style of questions as will the relative emphasis/weighting of the different topics.

25% Continuous Assessment

- Practical work based on 10 python practicals and 2+ online class tests[‡].
- In theory[§], weekly assignments, are graded in advance of next week.

[§]In practice, I ~~may~~ will fall behind a bit.

Who?

Dr Denis Flynn

📠 Room 327
☎ 051-30 2068
✉ Denis.Flynn@setu.ie



Background

- PhD in Applied Mathematics and Civil Engineering.
- BSc (H) Physics.

Academic Interests

- Dynamical systems, in particular systems with hysteresis
- Game development
- Languages: C/C++, Python, Java

Dr Kieran Murphy

📠 Room 313
☎ 051-30 2055
✉ Kieran.Murphy@setu.ie



Background

- PhD in Applied Mathematical Sciences
- BSc (H) Applied Mathematics.

Academic Interests

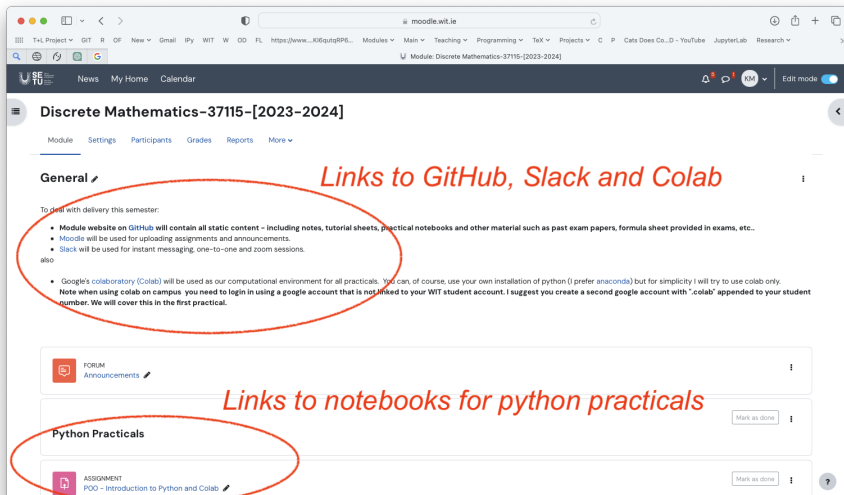
- Dynamical systems, in particular numerical analysis
- Game development
- Languages: C/C++, Python, Java

Outline

1. Module Introduction	2
1.1. What is the Aim of this Module?	3
1.2. Why Study Discrete Mathematics?	12
1.3. How will the Module be Delivered? Assessed?	14
1.4. Who is delivering this module?	17
1.5. When will the module be delivered?	18
2. Resources	18
2.1. Moodle	19
2.2. Github	20
2.3. Slack	21
2.4. Colab	22
2.5. PyTutor	23
3. Final Comments	25

Resources — Moodle

- URL: moodle.wit.ie/course/view.php?id=226896
- Used for all notices, assignment and practical work submissions.



The screenshot shows the Moodle course page for Discrete Mathematics-37115-[2023-2024]. The page has a dark header with navigation links and a sidebar. The main content area is titled 'Discrete Mathematics-37115-[2023-2024]' and includes a 'General' section. A red circle highlights the 'General' section, which contains links to GitHub, Slack, and Colab. Another red circle highlights the 'Python Practicals' section, which contains a link to notebooks for python practicals.

Discrete Mathematics-37115-[2023-2024]

Module Settings Participants Grades Reports More

General

To deal with delivery this semester:

- Module website on [GitHub](#) will contain all static content – including notes, tutorial sheets, practical notebooks and other material such as past exam papers, formula sheet provided in exams, etc..
- Moodle will be used for uploading assignments and announcements.
- Slack will be used for instant messaging, one-to-one and zoom sessions.

also

- Google's [collaboratory \(Colab\)](#) will be used as our computational environment for all practicals. You can, of course, use your own installation of python (I prefer [anaconda](#)) but for simplicity I will try to use colab only. Note when using colab on campus you need to login in using a google account that is not linked to your WIT student account. I suggest you create a second google account with ".colab" appended to your student number. We will cover this in the first practical.

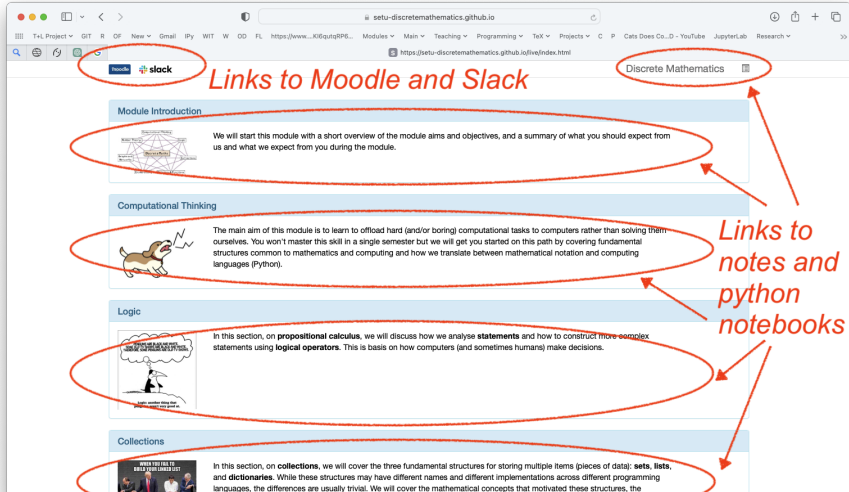
FORUM
Announcements

Python Practicals

ASSIGNMENT
POO – Introduction to Python and Colab

Resources — Github

- URL: [SETU-DiscreteMathematics.github.io/live](https://setu-discretemathematics.github.io/live)
- Used for all content (slides, notebooks, tutorial sheets).



The screenshot shows the website [setu-discretemathematics.github.io](https://setu-discretemathematics.github.io/live) in a browser. The browser's address bar and tabs are visible at the top. The website content is organized into sections, each with a blue header. Red circles and arrows highlight specific elements:

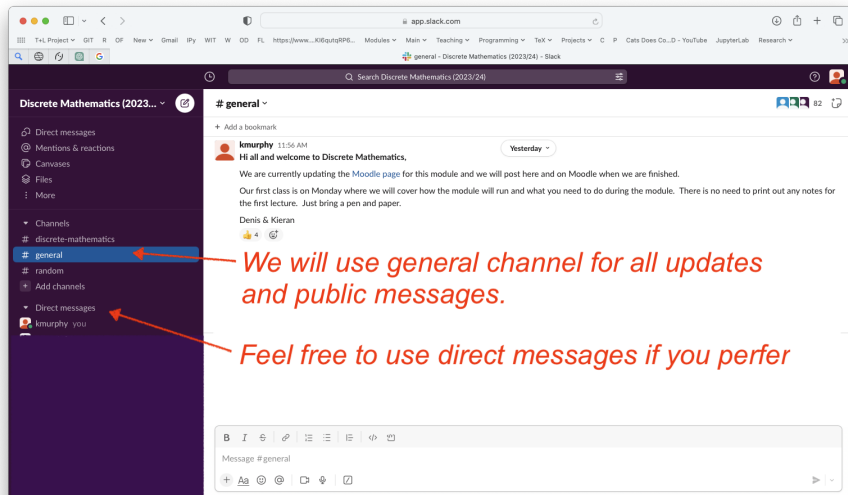
- A red circle around the Slack logo in the top navigation bar.
- A red circle around the "Discrete Mathematics" link in the top right corner.
- A red circle around the "Module Introduction" section header.
- A red circle around the text in the "Module Introduction" section: "We will start this module with a short overview of the module aims and objectives, and a summary of what you should expect from us and what we expect from you during the module."
- A red circle around the "Computational Thinking" section header.
- A red circle around the text in the "Computational Thinking" section: "The main aim of this module is to learn to offload hard (and/or boring) computational tasks to computers rather than solving them ourselves. You won't master this skill in a single semester but we will get you started on this path by covering fundamental structures common to mathematics and computing and how we translate between mathematical notation and computing languages (Python)." An illustration of a dog is also present.
- A red circle around the "Logic" section header.
- A red circle around the text in the "Logic" section: "In this section, on **propositional calculus**, we will discuss how we analyse **statements** and how to construct more complex statements using **logical operators**. This is basis on how computers (and sometimes humans) make decisions." An illustration of a person thinking is also present.
- A red circle around the "Collections" section header.
- A red circle around the text in the "Collections" section: "In this section, on **collections**, we will cover the three fundamental structures for storing multiple items (pieces of data): **sets**, **lists**, and **dictionaries**. While these structures may have different names and different implementations across different programming languages, the differences are usually trivial. We will cover the mathematical concepts that motivated these structures, the

Red text annotations with arrows point to the highlighted sections:

- "Links to Moodle and Slack" points to the Slack logo.
- "Links to notes and python notebooks" points to the "Discrete Mathematics" link and the "Module Introduction" section.

Resources — Slack

- URL: discretemathe-7co3349.slack.com
- Used for instant messaging, one-on-one sessions, etc.



Resources — Colab

- We will use the online Google Colab[¶] environment to code in python for all of our practical work.
- You can open a notebook from these slides by clicking the "OPEN in COLAB" icon or clicking/scanning the QR code



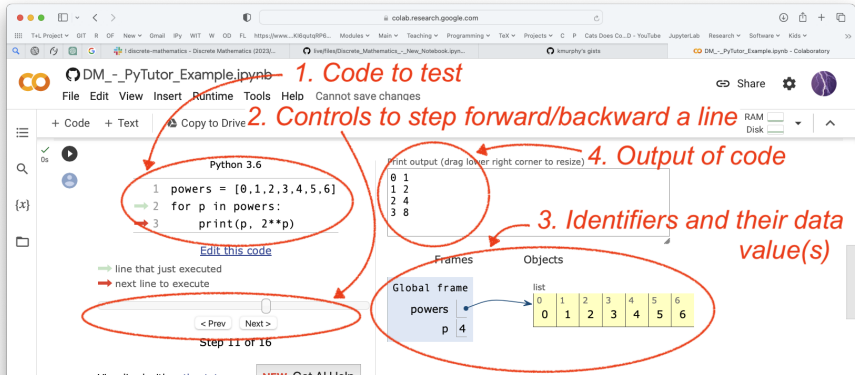
Since using Colab is such a large part of our module we have a separate set of slides covering using Colab in more detail.



[¶]Alternatively, if you want to install python on your laptop you could use the anaconda distribution from www.anaconda.com (just install the latest 64-bit, version 3.+).

Resources — PyTutor

- PyTutor (pythontutor.com) is a website that helps programmers to learn Python, Javascript, C/C++, and Java by visualising code execution (shows what happens to data as code runs, line by line).
- We will run PyTutor from within Colab so will demo PyTutor when we cover Colab (or click/scan the QR code).

The screenshot shows the PyTutor interface within a Google Colab notebook. The interface includes a code editor, a runtime control panel, and a visual representation of the program's state.

1. Code to test: The code being executed is:

```
Python 3.6
1 powers = [0,1,2,3,4,5,6]
2 for p in powers:
3     print(p, 2**p)
```

2. Controls to step forward/backward a line: The runtime control panel at the bottom left includes buttons for "< Prev" and "Next >".

3. Identifiers and their data value(s): The "Frames" section shows the current state of variables. In the "Global frame", the variable "powers" is a list, and "p" is the integer 4.

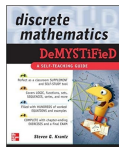
4. Output of code: The "Print output" section displays the results of the code execution:

```
0 1
1 2
2 4
3 8
```

The "Objects" section shows a visual representation of the list "powers" as a horizontal array of cells containing the values 0, 1, 2, 3, 4, 5, and 6.

Text Books

I like the following textbooks on discrete mathematics and expect that my notes will overlap significantly with these books. I do encourage you to read* them[†], however, be aware they may use different notation or cover different topics.



Discrete Mathematics Demystified

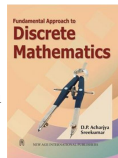
by Steven Krantz

Touches on nearly all of the topics that we hope to cover. We will probably go into greater depth in places, but a very nice and short read.

Fundamental Approach to Discrete Mathematics

by D. P. Acharjya Sreekumar

I also liked this book, however, due to time constraints, this module only focuses on material in chapter 1–4, 8, and 10.



*or skim them over a coffee or two.

[†]I also like *Applied Discrete Structures* by Alan Doerr and Kenneth Levasseur — it is a good source of exercises. (and is free (legally))

Outline

1. Module Introduction	2
1.1. What is the Aim of this Module?	3
1.2. Why Study Discrete Mathematics?	12
1.3. How will the Module be Delivered? Assessed?	14
1.4. Who is delivering this module?	17
1.5. When will the module be delivered?	18
2. Resources	18
2.1. Moodle	19
2.2. Github	20
2.3. Slack	21
2.4. Colab	22
2.5. PyTutor	23
3. Final Comments	25

Final Comments on Module

- Discrete Mathematics concepts appear either directly or indirectly in approximately 22 of the 30 modules on your degree.
 $\implies$ *Knowing Discrete Mathematics concepts greatly simplifies rest of the course.*
- The module is intended to be an introduction to a large number of topics, so treatment is broad rather than deep.
 - ✓ Most of material is at an introductory level.
 - ⚠ Keeping in sync with material, practicals and tutorials is important.
- The continuous assessment (the practicals) is intended to reinforce the connections between programming and discrete mathematics.

The CA is a “carrot not a stick” — we want you to enjoy the module and keep up to date with the material.

